

# S SETHU MADHAV

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## PROFESSIONAL SUMMARY

AI/ML Engineer with 1.5+ years of experience in building production-grade Computer Vision and Generative AI systems. Specializing in real-time inference, edge AI deployment, and autonomous systems. Delivered 80%+ inference latency reductions across production CV pipelines face recognition, object detection (YOLOv8/v26/SSD), pose estimation, and gesture recognition via ONNX, NCNN, Apache TVM, and INT8 quantization on Raspberry Pi-class hardware. Built autonomous indoor navigation achieving 75%+ reliability using monocular visual odometry, and extended into multimodal AI by integrating on-device LLMs (TinyLlama, Llama.cpp, Phi-2) with vision and speech pipelines for fully offline operation at sub-2s response latency. Experienced in robotics stack development (ROS2, SLAM, sensor fusion) and cloud-to-edge deployment via AWS SageMaker and Bedrock.

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, SQL, Bash, Shell Scripting, CMake.

**AI / ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, LLMs, RAG, FAISS, TinyML, Llama.cpp.

**Computer Vision:** OpenCV, YOLO, SSD, MobileNet, Face Recognition, Gesture Recognition, Pose Estimation, Visual Odometry.

**Model Optimization:** ONNX, NCNN, Apache TVM, TensorRT, INT8 Quantization, Pruning, Knowledge Distillation, TFLite.

**Embedded / Robotics:** Raspberry Pi, ESP32, Arduino, ROS2, SLAM, Sensor Fusion, Path Planning, I2C/SPI/UART/MQTT.

**Cloud / DevOps:** AWS EC2, SageMaker, Bedrock, Lambda, API Gateway, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD.

**Video Analytics / Streaming:** RTSP, WebRTC, GStreamer, FFmpeg.

## PROFESSIONAL EXPERIENCE

**Vintillix Global Innovation Pvt. Ltd,** Hyderabad, India

**AI/ML Engineer**

**September 2024-Present**

- Deployed optimized CNN models for face recognition, object detection, tracking, and segmentation (YOLOv8/SSD) across edge devices, maintaining 80%+ mAP accuracy under strict CPU/RAM constraints with no GPU dependency.
- Engineered real-time computer vision pipelines in C++ and Python on Raspberry Pi, reduced inference latency by 80%+ through hardware model compression using ONNX, NCNN, TensorRT, and Apache TVM frameworks.
- Built an autonomous indoor navigation stack using Raspberry Pi and monocular USB camera, implemented visual odometry, obstacle avoidance, and path planning achieving 70%+ reliability in dynamic indoor environments.
- Optimized on-device LLMs (TinyLlama, Llama.cpp) for fully offline inference on Raspberry Pi via quantization, pruning, and knowledge distillation, reducing model memory footprint by 60%+ with under 2-3% accuracy loss.
- Integrated optimized on-device LLMs (Phi-2, Gemma, Llama) with computer vision and speech pipelines to build a fully offline multimodal assistant on Raspberry Pi, achieving sub-2s end-to-end response latency across gesture, face, and voice interaction with zero internet dependency.
- Managed GPU-based model training and serving across SageMaker, Bedrock, EC2, and S3 via Docker and CI/CD pipelines, supporting reproducible deployments across heterogeneous hardware targets with 90%+ pipeline success rate.

**QuantumEco,** Chennai, India

**Software Developer**

**2024**

- Developed scalable Java/Spring Boot microservices with multithreading, OAuth2/JWT security, Docker/Kubernetes deployment, and CI/CD pipelines on AWS event-driven architecture.

## PROJECTS

**Face Recognition System** | OpenCV, ONNX, INT8 Quantization

- Designed and deployed a real-time face detection and recognition pipeline on Raspberry Pi using a custom mathematical feature extractor; applied quantization techniques to reduce memory footprint by 60% while maintaining high recognition accuracy at 20+ FPS.
- Implemented real-time face tracking using Kalman filters and integrated a PID controller for smooth servo-driven head movement on a humanoid robot, achieving stable target lock with sub-100ms control loop latency across dynamic environments.

**Visual Navigation - Monocular Camera** | Monocular camera, OpenCV, Visual Odometry, Path Planning

- Implemented fully camera-based autonomous indoor navigation (obstacle avoidance and path planning) without LiDAR or depth sensors using monocular vision, Kalman filtering, and dead reckoning. Achieved 80%+ navigation reliability with a real-time C++ perception-to-motion pipeline running at 8–10 FPS under 300 ms latency.

### **Object Detection & Recognition** | YOLOv8, SSD, Apache TVM, ROS2

- Built multi-class real-time object detection pipeline using MobileNetV2/YOLO + SSD, compiled with Apache TVM for ARM hardware, achieved 15+ FPS at 640×480 resolution on Raspberry Pi 4 integrated with Visual camera Navigation for obstacle avoidance.

### **Gesture & Pose Recognition** | MediaPipe, PyTorch, ONNX Runtime, NCNN

- Developed custom gesture/action classification and full-body pose estimation model, engineered touchless robot control interface achieving sub-500ms end-to-end latency with robust generalization across varied lighting and backgrounds.

## **EDUCATION**

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**Gokaraju Rangaraju Institute of Engineering & Technology**, Hyderabad, India

***B.Tech, Electronics and Communication Engineering***

**2024**